

MOTIONLAB

Unified Product Requirements Document

Train with Understanding

Version 1.0 · June 2026 · Confidential

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1. Executive Summary

1.1 Vision

MotionLab is the first platform to unify sports science education, intelligent workout planning, and sport-specific performance tracking in a single coherent experience. Athletes learn how to move correctly, train with AI-generated plans that adapt to their progress, and understand why every exercise, warmup, and recovery protocol exists — grounded in biomechanics and reviewed by sports science experts.

1.2 Problem Statement

Athletes at every level face the same structural gap: the knowledge that prevents injuries and unlocks performance is locked inside physio clinics and university sports science departments that most people never access. The existing landscape is fragmented:

- YouTube provides technique videos but no structure, personalisation, or injury context
- Fitness apps (MyFitnessPal, Hevy, StrongApp) track workouts but do not teach movement
- Sports physios provide expert guidance but cost ₹2,000–5,000 per session and are reactive, not preventative
- General fitness platforms ignore sport-specific biomechanics entirely
- No platform connects what a user learns about movement science to how they actually train day-to-day

The result: recreational athletes sustain preventable injuries, underperform their potential, and never develop a true understanding of how their body works during sport.

1.3 The Unified Solution

MotionLab merges two complementary MVPs — MotionLab (sports science education + community) and KavaFit (AI workout planning + live session tracking + nutrition) — into one platform where:

- Education informs training: movement science content grounds every exercise recommendation
- Training generates insight: session data surfaces relevant injury prevention and recovery content
- Community compounds learning: expert-verified threads and sport-tagged discussions extend the platform beyond content into lived experience
- AI connects everything: a context-aware coach with full training history, nutrition, movement science, and sport profile answers the questions a physio would answer

1.4 Goals

Product Goals

- Ship a production-ready MVP within 12 weeks using KavaFit's codebase as the foundation
- Launch with at least 2 sports fully covered (Table Tennis, Football) with expert-reviewed content
- Achieve Day-30 retention above 35% within 6 months of launch
- Reach 1,000 weekly active users within 3 months

Business Goals

- Establish product-market fit with fitness-committed recreational athletes aged 18–40 in India
- Build a freemium model gated on AI coach usage, advanced learning paths, and certifications
- Create the foundation for a B2B play: sports academies, college teams, corporate wellness
- Develop an expert contributor network of 5+ credentialed physios and sports scientists within 6 months

1.5 Non-Goals (v1.0)

- Native iOS or Android apps — v1.0 is a progressive web app
- Elite competitive athlete periodisation — clinical positioning is out of scope

- Team-based or group training coordination
- Wearable integrations (Apple Watch, Garmin, Fitbit)
- Real-time multi-device synchronisation
- In-app payment processing — subscription handled externally in v1.0
- Clinical or medical claims — MotionLab is not a medical device

1.6 Success Metrics

Metric	Target	Timeframe
Weekly Active Users (WAU)	1,000+	Month 3
Day-30 retention	>35%	Month 6
Session completion rate	>65%	Month 3
AI plan acceptance rate	>70%	Month 2
Avg sessions logged per user/week	2.5+	Month 3
Expert content pieces published	20+	Month 4
Community posts per week	50+	Month 3
NPS score	45+	Month 6

2. Product Analysis

This section documents the two existing products that were merged to form MotionLab. It provides a complete feature inventory, comparative analysis, and the rationale for every merge, keep, and discard decision.

2.1 MotionLab (Product A)

Overview

MotionLab is a sports science education platform built in Lovable, targeting recreational athletes who want to understand movement science, injury prevention, and biomechanics for their specific sports. It was founded by Aditya Saiprasad, whose personal experience of a scapular injury and physiotherapy rehabilitation revealed a structural gap: most athletes never receive the foundational sports science knowledge that physiotherapists take for granted.

Architecture

- Frontend: React generated via Lovable (no clean TypeScript architecture)
- Backend: Supabase (Auth, PostgreSQL, Storage, RLS)
- AI: Claude/Lovable AI for 4 sport-specific coaches with per-user chat history
- Community: Supabase-backed discussion board with sport tags and topic filters
- Analytics: None wired at MVP stage
- Error tracking: None wired at MVP stage

Feature Inventory

Feature	Status	Quality	Keep / Action
Landing page	Live	Good	Keep, redesign in new codebase
About page (founder story)	Live	Good	Keep — strong credibility asset
Sport explorer (8 sports)	Live	Good	Keep, extend
Sport detail pages (5 pillars)	Live	Good	Keep, extend
Movement science viewer	Live	Strong differentiator	Keep — core moat
Injury prevention hub	Live	Good	Keep — core moat
Expert hub	Live	Partial	Keep, build out contributor system
Community (Reddit-style)	Live	Needs work	Keep, redesign UX
AI coaches (4)	Live	Partial	Migrate to KavaFit AI architecture
Learning paths	Mockup only	Not built	Build properly in new platform
Real lesson progress tracking	Not built	—	Build in new platform
Admin CMS	Not built	—	Critical — build in Phase 2
Global search	Not built	—	Build in Phase 2
Notifications	Not built	—	Build in Phase 2
Resource library	Partial	Thin	Build properly in new platform
Profile + My Library	Partial	Thin	Merge with KavaFit profile
Certifications	Not built	—	Phase 3
Gym finder	Not built	—	Use KavaFit's implementation

Strengths

- Founder-market fit is exceptional: Aditya is the target user
- Visual-first pedagogy (skeleton overlays, force diagrams) is genuinely differentiated
- Injury prevention + movement science angle has strong SEO and emotional resonance
- Community architecture (sport tags, expert verification) is well-conceived
- Real Supabase backend with RLS is production-ready

Weaknesses

- Lovable-generated codebase: not maintainable at scale, no clean TypeScript architecture
- No real lesson progress tracking wired to the database
- AI coaches not grounded in MotionLab content — give generic answers
- Community UX is too rigid: no image uploads, no upvotes, poor post composer
- Zero analytics, error tracking, or observability
- No content pipeline: no Admin CMS means no way to add lessons without code changes

2.2 KavaFit (Product B)

Overview

KavaFit is an AI-powered fitness coaching SaaS delivering personalised workout planning, live session tracking with real-time PR detection, nutrition logging, and a contextual AI coach — all in a mobile-first web application. It targets fitness-committed individuals aged 22–42 who want scientifically grounded, adaptive training without the cost of a personal trainer. Its technical architecture is significantly more mature than MotionLab's.

Architecture

- Frontend: React 19 + TypeScript 6 + Tailwind CSS 4 + Vite 8 — production-grade
- Backend: Supabase (Auth, PostgreSQL, Edge Functions, Storage)
- AI: Groq LLM via Supabase Edge Function (ai-proxy) — server-side key management
- Offline: IndexedDB (kavafit-offline-v1) with background sync — real technical moat
- Analytics: PostHog (user behaviour)
- Error tracking: Sentry (runtime errors)
- Maps: Leaflet + OpenStreetMap (gym finder)
- Nutrition: Dual DB — Indian foods (pre-seeded) + USDA API (Edge Function)

Feature Inventory

Feature	Status	Quality	Keep / Action
AI workout planning (PPL, Full Body, Bro Split)	Live	Strong	Keep — core differentiator
Live session tracking + PR detection	Live	Strong	Keep — core differentiator
Progressive overload engine	Live	Strong	Keep — technical moat
40-group muscle taxonomy + volume tracking	Live	Strong	Keep — unique depth
Deload detection (5-week monitor)	Live	Strong	Keep — rare feature
Offline mode (IndexedDB + sync)	Live	Strong	Keep — real engineering moat
Nutrition logging (Indian food DB + USDA)	Live	Strong	Keep — strong India fit
AI recipe generation + grocery list	Live	Good	Keep
Body Lab (interactive anatomy)	Live	Good	Merge with MotionLab movement science
Dashboard (heatmap, streak, AI insights)	Live	Strong	Keep, extend with learning data
Progress tracking (measurements, photos)	Live	Good	Keep
Gym finder (GPS, OSM, Leaflet)	Live	Good	Keep
Excel export	Live	Good	Keep
Warm-up AI modal	Live	Good	Extend with sport-specific warmups from MotionLab content
Sentry + PostHog	Live	Good	Keep — use for merged product
Social features	Not built	—	Use MotionLab community
Education / learning content	Not built	—	Use MotionLab content system
Expert network	Not built	—	Use MotionLab expert hub

Strengths

- Production-grade React + TypeScript codebase — maintainable and extensible
- Groq AI architecture is mature: buildAgentContext() assembles full user context with token budget management

- Offline-first architecture is a genuine technical differentiator
- Indian food database is a strong localisation advantage
- PostHog + Sentry already wired — observability is production-ready
- 40-group muscle taxonomy is unique depth not found in competitor apps
- Progressive overload and deload detection show real sports science thinking

Weaknesses

- No education layer: tracks training but does not teach why
- No community: users have no social layer or peer learning
- AI answers are generic without sport-specific or movement science grounding
- Warmup generation is AI-only, not grounded in expert-reviewed content
- No sport-specific context: treats all users as gym athletes, not sport players

2.3 Comparative Analysis

Dimension	MotionLab	KavaFit	Decision
Codebase quality	Lovable-generated, not maintainable	Production React/TS, maintainable	Use KavaFit as base
AI architecture	Claude/Lovable, no context management	Groq, buildAgentContext(), Edge Function	Use KavaFit AI system
Workout tracking	None	Full: PR detection, progressive overload, volume	KavaFit wins — keep all
Sports science education	Full: movement, injury prevention, biomechanics	None	MotionLab wins — keep all
Nutrition	None	Full: Indian DB, USDA, AI recipes	KavaFit wins — keep all
Community	Reddit-style (needs UX work)	None	MotionLab — rebuild UX properly
Expert network	Designed, partially built	None	MotionLab — build out properly
Offline support	None	IndexedDB + background sync	KavaFit wins — keep all
Analytics + errors	None	PostHog + Sentry	KavaFit wins — keep all
Anatomy/body model	Skeleton overlays (education focus)	Body Lab 40 groups (training focus)	Merge into unified anatomy layer

Dimension	MotionLab	KavaFit	Decision
Indian market fit	Moderate	Strong (food DB, Bangalore persona)	KavaFit base, extend for sports
Dashboard	Education-focused, thin	Data-rich: heatmap, streak, nutrition	Merge: KavaFit data + ML learning widgets
Auth	Supabase (basic)	Supabase (Google OAuth + email, complete)	KavaFit auth system
Profile schema	Sports interests, learning goals	Goals, equipment, injuries, nutrition targets	Merge into one profiles table

3. User Personas

3.1 Primary Persona — The Sport-Active Amateur

Persona Profile

Age 22–35 · Metro India (Bangalore, Mumbai, Delhi) · Plays 1–2 recreational sports weekly · Trains at gym 2–4x/week · Has goals (performance, fitness, injury avoidance) but lacks structured sport-specific training · Willing to pay for something that actually works

Goals

- Improve performance in their sport without risking injury
- Understand why exercises and warmups matter — not just follow a plan blindly
- Get AI-level coaching without ₹5,000/session physiotherapy costs
- Find community with people who play the same sports

Pain Points

- Plays football or TT but has no idea if their movement mechanics are increasing injury risk
- Has experienced an injury (common: ACL, shoulder, wrist) that could have been prevented
- Generic fitness apps don't understand their sport — suggest bench press to a TT player
- YouTube gives technique tips but no personalised application or injury context

Representative User — Aditya

Plays competitive Table Tennis and recreational football. Suffered a scapular injury. Visited a sports physio who taught him warmup mechanics and biomechanics. Realised this knowledge should be accessible to everyone who plays sport, not just those who can afford physio.

3.2 Secondary Persona — The Returning Gym Athlete

Persona Profile

Age 28–42 · Career professional · Had a structured gym routine 3–5 years ago · Returning after a break · Injury history is relevant · Needs efficient, adaptive programming · Values nutrition tracking · Wants to understand muscle recovery and deload logic

Goals

- Get back to previous fitness levels efficiently
- Not re-injure themselves through poor programming
- Understand progressive overload and recovery — tired of guessing
- Track nutrition without spending 45 minutes logging a meal

3.3 Tertiary Persona — The Sports Science Learner

Persona Profile

Age 18–28 · College student or recent graduate · Interested in biomechanics and sports science · May not be an elite athlete but is intellectually curious · Attracted by the visual learning layer · Likely to share content and build community

Goals

- Learn sports science concepts in an accessible, visual format
- Build understanding they can apply to their own sport or share with others
- Connect with a community of similarly minded people

3.4 Expert Contributor Persona — The Sports Physio / Coach

Persona Profile

Age 26–45 · Sports physiotherapist, sports scientist, or coach · Wants to build audience and online presence · Creates content on Instagram or LinkedIn · Motivated by visibility and credibility, not payment at MVP stage · Values the Expert badge and verified contributor status

Goals

- Reach a wider audience of athletes who need their expertise
- Be attributed for content in a trusted, science-backed platform
- Respond to Ask-an-Expert threads to demonstrate knowledge

4. Information Architecture

4.1 Navigation Structure

Public Navigation (logged out)

Nav Item	Path	Purpose
Home	/	Marketing landing page
About	/about	Founder story — Aditya Saiprasad
Sports	/sports	Public sport explorer (preview)
Experts	/experts	Expert contributor profiles
Resources	/resources	Public articles and guides
Gym Finder	/gyms	GPS gym locator (public)
Contact	/contact	Contact form + FAQ
Login	/auth	Sign in
Get Started	/auth?mode=signup	Primary CTA

Authenticated Navigation — Train

Nav Item	Path	Purpose
Dashboard	/dashboard	Unified hub: training + learning
Workout	/workout	Plan management + live session tracking
Nutrition	/nutrition	Meal logging, macros, AI recipes
Progress	/progress	Volume trends, measurements, photos, analytics
Body Lab	/anatomy	Interactive 40-group muscle anatomy

Authenticated Navigation — Learn

Nav Item	Path	Purpose
Sports Library	/sports	All sports, locked/unlocked content
Learning Paths	/learn	Structured beginner/intermediate/advanced journeys
Movement Science	/movement-science	Interactive biomechanics viewer
Injury Prevention	/injury-prevention	Sport-specific injury prevention hub
Recovery	/recovery	Recovery protocols and load management

Authenticated Navigation — Connect

Nav Item	Path	Purpose
Community	/community	Reddit-style sport-tagged discussion
Expert Hub	/experts	Authenticated expert profiles + content
AI Coach	/ai	Full-screen AI coach with context
My Library	/library	Saved lessons, articles, training plans

Account

Nav Item	Path	Purpose
Profile	/profile	Personal info, sports, achievements, certifications
Settings	/settings	Account, security, notifications, privacy
Notifications	/notifications	In-app notification centre

Admin (role-gated)

Nav Item	Path	Purpose
Admin Overview	/admin	Platform analytics, user counts, content status
Sports	/admin/sports	Manage sports, pillars, metadata
Lessons	/admin/lessons	Create/edit/publish lesson content
Learning Paths	/admin/paths	Manage path structure and progression
Exercises	/admin/exercises	Exercise database management
Experts	/admin/experts	Expert contributor management
Resources	/admin/resources	Articles, guides, handbooks
Users	/admin/users	User management, roles, moderation
Analytics	/admin/analytics	Platform-wide usage analytics

5. Feature Specification

5.1 Onboarding

A unified 6-step wizard covering both fitness and sport dimensions. Replaces KavaFit's 5-step wizard and MotionLab's 3-step onboarding with a single coherent flow.

Step	Data Collected	Purpose
1. Profile	Name, age, weight, height	Baseline for calculations and personalisation
2. Fitness Goals	Goal (muscle gain / fat loss / general fitness), experience level, sessions/week	Plan structure + volume targets
3. Equipment	Available equipment, current injuries	Exercise filtering + safety constraints
4. Sports	Sports played, frequency, competition level	Sport-specific warmup, content, community
5. Nutrition	Calorie target, protein target, dietary preference, allergies	Macro targets + food DB filtering
6. Plan Generation	AI generates first workout plan + sport warmup protocol	First-session readiness

Acceptance Criteria

- Onboarding completes in under 4 minutes for the average user
- All fields from both KavaFit and MotionLab onboardings are captured
- AI workout plan is generated and ready before user reaches Dashboard
- Sport-specific content is surfaced immediately after onboarding based on sports selected
- Progress is saved after each step — closing and reopening resumes from last completed step

5.2 Dashboard

A unified hub that merges KavaFit's data-rich dashboard (heatmap, streak, nutrition, AI insights) with MotionLab's learning progress. The user should see both their training state and their learning state in one view.

Required Widgets

- Personalised greeting with name + training streak count
- Today's scheduled workout preview with one-tap Start
- Nutrition progress bars: calories and protein vs daily target
- Muscle heatmap: current week volume by muscle group (40-group taxonomy)
- AI insights card: deload alerts, low-volume warnings, PR celebrations, streak milestones (24h cache)
- Continue Learning card: next lesson in active learning path
- Injury Prevention card: contextual warning based on recent training volume (e.g. high hamstring volume → hamstring care module)
- Sport Schedule widget: user enters their next sport session → triggers warmup reminder flow
- Weekly AI summary: generated each Monday, cached in localStorage

Priority

Must Have

5.3 AI Workout Planning

Inherited from KavaFit with sport-specific extensions. Generates structured weekly split plans using profile, goals, equipment, injuries, and — new in the merged product — sport schedule.

Supported Plan Templates

- Push / Pull / Legs (PPL)
- PPL + Upper / Lower hybrid
- Full Body
- Bro Split
- Sport-Supplementary (designed around a user's sport schedule — gym sessions fill recovery days)
- One-off single session generation

Sport-Supplementary Planning (New)

When a user specifies sport days (e.g. football Tuesday + Saturday), the AI generates gym sessions that:

- Avoid heavy leg work the day before sport sessions
- Prioritise sport-relevant strength (hip flexor strength for football, wrist stability for TT)
- Build in appropriate recovery between gym and sport

5.4 Live Session Tracking

Inherited from KavaFit unchanged. This is one of the strongest features in either product.

Feature	Behaviour	Priority
Set logging	Reps, weight, RPE per set with optimistic UI	Must
PR detection	Compares to historical bests, shows PR badge, 24h cache	Must
Progressive overload	Compound +5kg / Isolation +2.5kg suggestion	Must
Sport warmup integration	Pre-session: user selects sport → AI generates sport-specific warmup from MotionLab content	Must
Gym warmup	AI-generated 5-exercise warmup modal	Must
Rest timer	Configurable countdown	Must
Offline mode	IndexedDB queue, background sync, idempotent writes	Must
RPE guidance	Tooltip explaining RPE scale contextually	Should
Session notes	Free-text field on session completion	Should

5.5 Sport-Specific Warmup Flow (New — Key Differentiator)

This is the feature that makes the merger feel intentional rather than additive. It connects MotionLab's education layer to KavaFit's tracking layer.

Flow

1. User sets sport session in Dashboard (e.g. Football — Saturday 10am)
2. Friday evening: push notification / in-app reminder with sport-specific warmup
3. User opens warmup: content sourced from MotionLab's injury prevention and movement science modules for that sport
4. Warmup is logged as a pre-session activity
5. After sport session, Recovery module for that sport is surfaced
6. If repeated high-volume sport weeks, injury risk flag surfaces relevant prevention content

Priority

Must Have — this is the MVP's strongest differentiator

5.6 Nutrition

Inherited from KavaFit unchanged. Indian food database is a genuine competitive advantage in the primary market.

- Dual food database: pre-seeded Indian foods + live USDA API lookup
- Daily macro tracking: calories, protein, carbohydrates, fat vs user targets
- Meal logging with per-ingredient macro breakdown
- AI recipe generation: Groq generates recipes with full macro breakdown
- Grocery list builder: AI suggests shopping from goals and recent meal history
- Sport nutrition guidance (new): post-sport recovery nutrition recommendations based on sport played and session intensity

5.7 Sports Science Education

MotionLab's core content system. Every sport gets 5 pillars. Content is expert-reviewed and visual-first.

Sport Pillars

Pillar	Content Types	Key Differentiator
Learn the Sport	Rules, history, tactics, equipment guides	Structured, not scattered like YouTube
Movement Science	Biomechanics lessons with skeleton overlays, force vectors, muscle activation	Visual-first — unique in market
Injury Prevention	Sport-specific common injuries, prevention exercises, warmup protocols	Physio-reviewed — credibility moat
Training	Sport-specific strength, mobility, conditioning programs	Connects to KavaFit workout system
Recovery	Sleep, hydration, load management, return-to-play protocols	Connects to KavaFit deload detection

Supported Sports at Launch

- Table Tennis (Aditya's primary sport — first and deepest coverage)
- Football / Soccer (Aditya's second sport)
- Tennis, Basketball, Badminton, Running, Cycling, Swimming (content added incrementally)

Content Quality Requirements

- All biomechanics content reviewed by a credentialled sports scientist or physiotherapist before publish
- Injury prevention content must include a disclaimer that it is not medical advice
- Visual diagrams must accurately represent anatomical structures — no generic stock illustrations

5.8 Learning Paths

Structured educational journeys per sport. Unlike learning a single lesson, a path takes a user from beginner to advanced with clear progression milestones.

Structure

- Each sport has 3 paths: Beginner, Intermediate, Advanced
- Each path contains 4–8 modules
- Each module contains 3–6 lessons
- Lessons are text + visuals + short exercises — 5–10 minutes each
- Completion is tracked in `lesson_progress` table and shown on Dashboard
- Completing a path unlocks a digital certificate

5.9 Community

Reddit-style discussion with sport-tagging and expert verification. Rebuilt from MotionLab's existing community with improved UX.

Post Composer

- Three post types: Text (rich text), Image (multi-image, Supabase Storage), Link (auto-fetch preview)
- Required: sport tag (from list of supported sports)
- Optional flair: Technique · Injury Question · Progress · Recovery · General

Feed

- Sort: Hot · New · Top this week
- Filter: by sport tag, by flair
- Upvote / downvote (one vote per user per post, toggleable)
- Nested comments (1 level deep at MVP)
- User avatar + display name prominent on every post

Expert Features

- Expert badge on verified contributor profiles and their replies
- Ask an Expert post type: elevated visual treatment, navy border, routed to relevant expert
- Experts can be tagged in posts

Moderation

- Admin can remove posts and comments
- Report button on all posts and comments
- User blocking (Phase 2)

5.10 AI Coach

The merged product's AI coach inherits KavaFit's Groq architecture and extends buildAgentContext() with MotionLab data.

Context Assembled Per Request

- User profile: age, weight, goals, equipment, injury history, sports
- Last 7 training sessions (10 sets each, truncated to 3 if over token budget)
- Weekly muscle volume by group
- Today's nutrition
- Today's scheduled plan
- Active learning path and last completed lesson
- Sport schedule for the week
- Recent injury flags from lesson interactions

AI Modes

Mode	Description	New / Existing
default	Free-form coaching conversation with full context	KavaFit
flags	Structured JSON: insights, alerts, celebrations for Dashboard	KavaFit
recipe	AI meal generation with macros as JSON	KavaFit
workout	Full session plan as structured JSON	KavaFit
warmup	Gym warmup sequence (5 exercises)	KavaFit
grocery	Shopping list from nutrition goals	KavaFit

Mode	Description	New / Existing
sport_warmup	Sport-specific warmup sourced from MotionLab injury prevention content	New
injury_check	Flag movement risk based on training patterns, surface relevant MotionLab module	New
learning_rec	Next lesson recommendation based on training data and learning progress	New

AI Grounding (Critical)

The three new AI modes must surface MotionLab lesson content, not generic answers. Implementation: relevant lessons are retrieved by sport + topic tag and included in the prompt context. This is the feature that transforms the AI from a generic chatbot into a platform-native coach.

5.11 Body Lab + Movement Science (Merged)

KavaFit's Body Lab (40-group muscle taxonomy, recovery guidance, exercise associations) and MotionLab's Movement Science viewer (skeleton overlays, force vectors, motion breakdowns) are merged into a unified anatomy and movement learning layer.

- Interactive body model: tap a muscle group to see volume, recovery state, and associated exercises (KavaFit)
- Movement overlays: for each sport, view skeleton overlay, muscle activation sequence, force direction (MotionLab)
- Recovery time guidance per muscle group (KavaFit)
- Injury risk indicators: muscle groups with high recent volume get a visual warning (new — connects both systems)

5.12 Progress Tracking

Inherited from KavaFit with learning analytics added.

- Calendar view: sessions logged per day, monthly overview
- Volume trend graphs: per-exercise volume over time
- Body measurements: weight, chest, waist, hips, arms, thighs
- Progress photos: upload, date-stamp, compare
- Learning analytics (new): hours learned, lessons completed, weekly activity, learning trends

- Certification tracker: completed paths and earned certificates
- Excel export: full session, measurement, and progress history

5.13 Admin CMS

Critical missing feature from both products. Without it, content cannot be added without code changes. Must be built in Phase 2.

Module	Capabilities
Sports Management	Add/edit sports, manage sport metadata, enable/disable content pillars
Lesson Management	Create/edit/publish lessons, upload visuals, manage content order
Learning Path Management	Define path structure, module order, completion criteria
Exercise Database	Add/edit exercises, muscle group mapping, difficulty, instructions
Expert Management	Invite experts, manage verification badges, link to content
Resource Publishing	Create/edit articles, guides, handbooks with rich text editor
User Management	View users, assign roles (admin/expert/user), handle reports
Platform Analytics	WAU, retention, content engagement, AI usage, community activity

6. User Flows

6.1 Onboarding Flow

7. User arrives at landing page → clicks "Get Started"
8. Sign up: email + password or Google OAuth
9. Step 1 — Profile: name, age, weight, height
10. Step 2 — Fitness Goals: goal, experience level, sessions/week
11. Step 3 — Equipment: available equipment, current injuries
12. Step 4 — Sports: select sports played, frequency (can skip if gym-only)
13. Step 5 — Nutrition: calorie target, protein target, dietary preference, allergies
14. Step 6 — Plan Generation: AI generates first plan + sport warmup protocol (loading state)
15. Land on Dashboard: plan ready, first lesson recommended, sport schedule widget empty

6.2 The Killer Loop — Sport Session Flow

This flow is the product's primary differentiator. It is the reason both products needed each other.

16. User enters sport session in Dashboard widget: "Football — Saturday 10am"
17. Friday 8pm: push notification "Your football warmup for tomorrow is ready"
18. User opens warmup: 8-minute football-specific warmup from MotionLab injury prevention content
19. After warmup, user marks it complete — logs to session history
20. Saturday: plays football session
21. Saturday afternoon: notification "Time to recover — here's your post-football recovery protocol"
22. Sunday: Dashboard shows injury risk flag if consecutive high-load football weeks
23. Flag links directly to relevant MotionLab injury prevention module

6.3 AI Coach Interaction

24. User opens AI Coach (full screen or floating drawer from any page)
25. buildAgentContext() assembles: profile + last 7 sessions + volume + nutrition + plan + learning progress + sport schedule
26. User asks: "Why does my shoulder hurt when I play TT?"
27. AI detects: shoulder question + TT context → triggers injury_check mode
28. injury_check mode retrieves relevant MotionLab lesson: "TT Shoulder Health" from injury prevention module
29. AI response: explains biomechanical reason + surfaces the lesson as an in-chat card
30. User taps the lesson card → taken to MotionLab lesson
31. Lesson completion tracked in lesson_progress table

6.4 Community Flow

32. User browses community — public, no login required
33. Finds a thread about football ACL prevention → reads expert reply with Expert badge
34. Wants to post: prompted to sign in
35. Opens post composer: selects post type (Text / Image / Link)
36. Selects sport tag (Football) + optional flair (Injury Question)
37. Writes post, optionally uploads image(s) to Supabase Storage
38. Post appears in feed with their avatar, name, sport tag, flair
39. Other users upvote and comment — nested one level
40. If post tagged as Ask an Expert, gets elevated treatment and routed to expert notification

7. Technical Architecture

7.1 Recommended Stack

Layer	Technology	Rationale
Frontend framework	React 19 + TypeScript 6	KavaFit's proven stack — maintainable, type-safe
Styling	Tailwind CSS 4	MotionLab design tokens applied as Tailwind theme config
Build tool	Vite 8	Fast HMR, optimal bundle sizes
Routing	React Router 7	KavaFit's proven setup
State management	React Context (Auth, Workout, UI)	Sufficient at MVP scale — no Redux needed
Backend	Supabase	Auth, PostgreSQL, Edge Functions, Storage, RLS — both products already use it
AI provider	Groq (primary)	Mature, fast inference, server-side via Edge Function — KavaFit architecture
AI fallback	Anthropic Claude (secondary)	For complex reasoning tasks where Groq underperforms
Offline	IndexedDB (kavafit-offline-v1)	KavaFit's proven offline architecture — real technical moat
Maps	Leaflet + OpenStreetMap	KavaFit gym finder — keep as-is
Analytics	PostHog	KavaFit's setup — keep as-is, extend with MotionLab events
Error tracking	Sentry	KavaFit's setup — keep as-is
Nutrition DB	Indian foods (pre-seeded) + USDA API (Edge Function)	KavaFit advantage — keep as-is

7.2 Frontend Architecture

Design Token System

MotionLab's colour palette applied as a Tailwind CSS theme extension:

- primary-green: #264653 (Forest Green)
- primary-olive: #606C38 (Olive Green)

- primary-navy: #1D3557 (Deep Navy Blue)
- accent-slate: #4A6FA5 (Slate Blue)
- accent-maroon: #6D071A (Deep Maroon — limited use)
- bg-light: #F3F4F6 (Cool Light Grey)
- bg-mid: #E5E7EB (Mid Grey)
- bg-dark: #1F2937 (Soft Charcoal Grey)
- surface: #FAFAFA (Off-White Grey)

Responsive Behaviour

- Large Desktop (1400px+): full navigation bar with all links visible
- Below 1400px: hamburger navigation with slide-out drawer
- Tablet: hamburger navigation
- Mobile: hamburger navigation, bottom-positioned CTAs, touch-friendly tap targets
- Maximum content width: 1440px, centred on all screen sizes

React Context Architecture

- AuthContext: user, session, profile, avatarUrl, loading, isAdmin
- WorkoutContext: workoutUpdate, activeSessionExercises, heatmapRefreshKey
- UIContext: drawerOpen, drawerInitMessage, open/close handlers
- LearningContext (new): activeLearningPath, lessonProgress, sportSchedule

7.3 AI Architecture

Request Flow

41. Client calls `callAgent(userId, message, mode?)` in `lib/agent.ts`
42. `buildAgentContext(userId)` assembles full context — capped at ~6,000 tokens
43. If mode requires MotionLab content: `retrieveRelevantContent(sport, topic)` fetches indexed lessons
44. Request proxied through Supabase Edge Function (ai-proxy) — `GROQ_API_KEY` never exposed to client
45. Groq LLM generates response
46. `parseAgentJSON()` handles structured mode responses with graceful fallback
47. Response streamed to client

Context Budget Management

Context Component	Token Budget	Truncation Rule
User profile	~300 tokens	Never truncated

Context Component	Token Budget	Truncation Rule
Last N training sessions	~2,000 tokens	Reduce from 7 → 5 → 3 sessions if over budget
Weekly volume by muscle group	~400 tokens	Never truncated
Today's nutrition	~300 tokens	Never truncated
Today's plan	~500 tokens	Truncate exercise list if needed
Learning path + last lesson	~300 tokens	Summary only if over budget
Sport schedule	~200 tokens	Never truncated
Retrieved lesson content (RAG)	~800 tokens	Top 1 result only if over budget
Total cap	~6,000 tokens	Hard cap — truncate in reverse priority order

Content Retrieval (RAG)

For injury_check and learning_rec modes, relevant MotionLab lessons are retrieved and included in the prompt context:

- Lessons are indexed by sport + topic tags in the lessons table
- At MVP: simple keyword + tag matching — no vector embeddings required
- Phase 3: add pgvector embeddings on lesson content for semantic search
- Retrieved lesson summary (not full content) is included in prompt — links to full lesson in response

7.4 Offline Architecture

Inherited from KavaFit unchanged. This is a genuine engineering moat.

- IndexedDB store: kavafit-offline-v1 with two object stores (pending-sets, pending-sessions)
- On reconnection: sync layer pushes all pending records to Supabase in order
- Writes are idempotent — duplicate sync attempts do not create duplicate records
- Session set logging works within 200ms of tap with zero connectivity
- Offline scope: workout tracking only — community and learning require connectivity

8. Database Design

Single Supabase PostgreSQL instance. All tables use Row Level Security (RLS). All user data tables enforce user-scoped access. Admin tables are role-gated.

8.1 Core Tables

profiles (merged)

Combines KavaFit's profiles table (fitness fields) with MotionLab's user data (sport and learning fields).

Column	Type	Source	Notes
id	uuid PK	Both	References auth.users(id)
name	text	Both	
email	text	Both	
avatar_url	text	KavaFit	Supabase Storage URL
age	int	KavaFit	
weight_kg	numeric	KavaFit	
height_cm	numeric	KavaFit	
fitness_goal	text	KavaFit	muscle_gain fat_loss general_fitness
experience_level	text	Both	beginner intermediate advanced
sessions_per_week	int	KavaFit	
equipment	text[]	KavaFit	Array of available equipment
injuries	text	KavaFit	Free text injury history
calorie_target	int	KavaFit	
protein_target	int	KavaFit	
dietary_preference	text	KavaFit	
sports	text[]	MotionLab	Array of sport slugs
sport_frequency	jsonb	New	{"football": 2, "table_tennis": 3}
learning_goals	text[]	MotionLab	
onboarding_complete	boolean	Both	Gates dashboard access
deload_suggested_at	timestamptz	KavaFit	
created_at	timestamptz	Both	

8.2 Training Tables (KavaFit — preserve as-is)

- exercises: ~400 pre-seeded, muscle groups, equipment, difficulty, instructions
- workout_plans: AI-generated or manual plans per user
- plan_days: day-level structure within a plan
- sessions: completed sessions with date, duration, notes
- session_sets: individual sets with reps, weight, RPE, timestamp
- measurements: 7-dimension body measurements over time
- muscle_volume_log: weekly volume per muscle group (11 tracking groups)
- meal_history: logged meals with full macro breakdown
- progress_photos: uploads with date and notes
- indian_foods: pre-seeded Indian food DB with per-serving macros

8.3 Education Tables (new)

Table	Key Columns	Purpose
sports	id, name, slug, description, icon_url, active	Sport catalogue
learning_paths	id, sport_id, title, level (beginner intermediate advanced), description	Structured learning journeys
modules	id, learning_path_id, title, order, description	Groupings within a path
lessons	id, module_id, title, content_body, content_type, visual_url, sport_tags[], topic_tags[], duration_minutes, expert_id, published	Individual lessons with content
lesson_progress	id, user_id, lesson_id, completed, completion_date	Per-user lesson tracking
experts	id, name, title, bio, photo_url, credentials, specialisation[], verified, user_id	Expert contributor profiles
articles	id, title, body, author_id (expert_id), sport_tags[], topic_tags[], published_at	Resource library articles
certifications	id, user_id, learning_path_id, issued_at, certificate_url	Earned certificates
achievements	id, user_id, achievement_type, earned_at, metadata jsonb	Badges and milestones

8.4 Community Tables (MotionLab — rebuild)

Table	Key Columns	Purpose
discussions	id, user_id, sport_tag, flair, post_type (text image link), title, body, image_urls[], link_url, link_preview jsonb, upvotes, downvotes, created_at	Community posts
comments	id, discussion_id, parent_comment_id, user_id, body, upvotes, downvotes, created_at	Nested comments (1 level)
post_votes	id, user_id, discussion_id, vote_type (up down), created_at	Vote tracking (unique per user per post)
comment_votes	id, user_id, comment_id, vote_type (up down), created_at	Comment vote tracking

8.5 Platform Tables

Table	Key Columns	Purpose
user_roles	id, user_id, role (admin expert user)	Role-based access control
coach_conversations	id, user_id, title, created_at	AI coach chat sessions
coach_messages	id, conversation_id, role (user assistant), content, mode, created_at	Per-message AI history
notifications	id, user_id, type, title, body, read, action_url, created_at	In-app notification queue
bookmarks	id, user_id, content_type (lesson article plan), content_id, created_at	Saved content (My Library)
sport_schedules	id, user_id, sport, day_of_week, time, active	Weekly sport schedule for warmup flow

9. Security

9.1 Authentication

- Supabase Auth — email + password + Google OAuth
- Password leak protection enabled

- Auto-confirm email on signup (relaxed for MVP, add verification in Phase 2)
- JWT tokens managed by Supabase — never stored in localStorage
- Session expiry: 1 hour access token, 7-day refresh token

9.2 Authorisation

- Row Level Security (RLS) enforced on all user data tables
- Policy: users can only read and write their own data
- Community posts: public read, authenticated write
- Admin tables: admin role required
- Expert content: expert role required to publish
- GROQ_API_KEY: server-side Supabase secret only — never in client bundle
- USDA_API_KEY: server-side only

9.3 Data Protection

- No PII logged to PostHog — anonymous session IDs only
- No PII in Sentry — scrub user-identifying fields before logging
- Progress photos stored in Supabase Storage with private bucket + signed URLs
- Community image uploads: public bucket (user-consented for community sharing)
- Data export available to users (GDPR + DPDP compliance)
- Account deletion: purges all user data from all tables

9.4 Content Moderation

- Community posts: report button on all posts and comments
- Admin review queue for reported content
- Automated: no profanity filter at MVP — admin moderation only
- Expert content: manual review required before publish flag is set

10. Performance Goals

Metric	Target	Notes
First Contentful Paint (FCP)	<1.5s on 4G mobile	Critical for India market

Metric	Target	Notes
AI coach response time	<3s standard queries	<5s for plan generation
Session set logging	Optimistic UI — no perceptible delay	IndexedDB write < 50ms
Offline set logging	Works within 200ms of tap	Zero connectivity
Page navigation	<500ms between routes	React Router + code splitting
Image uploads (community)	<5s for 3 images at 4G	Compress before upload
Lesson content load	<2s per lesson page	CDN for visual assets
Database query P95	<200ms	Supabase + indexes on user_id, sport_tag, created_at

11. Development Roadmap

Built in milestones using KavaFit's codebase as the foundation. MotionLab's pages are rebuilt inside it with the new design system applied. No code from the Lovable-generated MotionLab codebase is carried over.

Phase 1 — Foundation (Weeks 1–4)

Goal

One unified codebase, one design system, one Supabase project, one auth flow. Both founders can ship to it from day one.

- Decision: agree brand name and finalise
- Apply MotionLab colour palette as Tailwind theme extension in KavaFit codebase
- Merge Supabase schemas: merge profiles tables, confirm single Supabase project
- Rebuild MotionLab's Home, About, and Sport Explorer pages in new codebase
- Unified auth and onboarding (6-step wizard covering fitness + sports)
- Extend buildAgentContext() with sports + learning fields
- Deploy to production domain

Deliverables

- Single deployable application with both products' existing features preserved
- Unified design system applied consistently
- One Supabase project with merged schema

Phase 2 — Core Platform (Weeks 5–10)

Goal

The education layer is live and connected to training. The killer loop (sport warmup flow) works end to end. Admin CMS allows content to be published without code changes.

- Sport-specific warmup flow (the killer loop) — end to end
- Lesson content system: publish first 5 TT lessons + 5 Football lessons
- Learning paths: beginner paths for TT and Football
- Admin CMS: sports, lessons, paths, experts, resources
- Rebuild community with Reddit-style UX (image uploads, upvotes, post types)
- Expand Expert Hub: invite first 2 expert contributors, wire expert badge
- AI modes: sport_warmup, injury_check, learning_rec
- AI content grounding: lesson retrieval wired into injury_check and learning_rec

Deliverables

- MotionLab education content live for 2 sports
- Sport warmup flow working end to end
- Community rebuilt with full Reddit UX
- Admin CMS operational for both founders to publish content

Phase 3 — AI and Depth (Weeks 11–16)

Goal

The AI coach feels platform-native, not generic. Learning paths have full coverage for 4 sports. Push notifications drive the habit loop.

- Push notifications: warmup reminders, streak alerts, Monday AI summary
- Extend lesson content to Tennis and Running
- Pgvector embeddings on lesson content for semantic AI retrieval
- Certifications and achievement system
- Global search across sports, lessons, experts, articles
- Notification centre (in-app)
- Profile completion: achievements display, certification badges

- Settings page: account, security, privacy, notification preferences

Phase 4 — Growth and Monetisation (Weeks 17–24)

Goal

Revenue model live. Platform ready for marketing push. B2B groundwork laid.

- Freemium feature gating: free tier (3 AI queries/day, basic content) vs Pro (unlimited AI, all paths, certifications)
- External subscription management integration
- Wearable integrations: Apple Health + Google Fit step/HR ingestion
- B2B pilot: approach 2–3 sports academies or college teams for team access
- Social layer: share PRs, share certification achievements
- Extend content: Basketball, Badminton, Cycling, Swimming
- Coach portal: trainers assign plans, review client sessions

12. Risk Register

Risk	Likelihood	Impact	Mitigation
Content quality: biomechanics errors damage credibility	Medium	High	All content expert-reviewed before publish. Start with 2 sports at depth rather than 8 at breadth. Disclaimer on all health-adjacent content.
AI answers generic rather than platform-grounded	High	High	Build lesson retrieval into AI modes early (Phase 2). Test AI responses against MotionLab content catalogue before launch.
Lovable codebase migration causes regressions	Medium	Medium	KavaFit codebase is the base — MotionLab pages rebuilt, not migrated. No Lovable code is carried over.
Expert contributor acquisition stalls	Medium	Medium	Start with 1 physio Aditya already knows from rehab. Use MotionLab traffic as leverage from month 2+.
YouTube comparison: "why pay when YouTube is free?"	High	High	Lean into structure + personalisation + community — things YouTube cannot offer. The sport warmup flow is the clearest example.

Risk	Likelihood	Impact	Mitigation
Two founders, limited hours	High	Medium	Clear ownership split: Aditya owns education + content + community. KavaFit founder owns training + nutrition + infrastructure. Milestones are independent.
Indian OSM gym data quality in Tier 2 cities	Medium	Low	Secondary data source planned for Phase 4. Not blocking for metro India launch.
Progressive overload increments not configurable	Low	Low	Phase 3 item. Current +5kg/+2.5kg defaults are fine for MVP.

13. UX Guidelines

13.1 Design Philosophy

MotionLab should feel like the intersection of Apple (premium minimalism), Linear (developer-grade precision), and Notion (editorial spaciousness) — applied to sports science. Every design decision must reinforce trust, expertise, and education. The platform should feel scientific without feeling cold.

- Minimal: no decorative elements without purpose
- Spacious: generous whitespace — content breathes
- Visual-first for education: diagrams, overlays, and motion breakdowns are central, not supplementary
- Data-rich for training: heatmaps, progress charts, and streaks give users a sense of progress at a glance
- Trust signals everywhere: expert badges, research citations, and founder story are not afterthoughts

13.2 Component Patterns

- Glassmorphism: used sparingly for cards that overlay content (e.g. movement science overlays)
- Rounded corners: consistent 8–12px radius on cards, larger on modals
- Shadows: subtle only — 0 0 24px rgba(0,0,0,0.06)
- Typography: large editorial headings (48–64px hero), clean sans-serif body, strong hierarchy

- Empty states: every empty state has an illustration, helpful message, and relevant CTA
- Loading states: skeleton loaders on all data-dependent components
- Error states: always explain what happened and what to do next — never raw error strings

13.3 Accessibility

- WCAG 2.1 AA compliance target for all interactive elements
- Keyboard navigation for all primary user flows
- Reduced motion respected via prefers-reduced-motion media query
- Minimum touch target size: 44px × 44px on mobile
- Colour contrast ratio: minimum 4.5:1 for normal text, 3:1 for large text

14. Future Enhancements

Intentionally deferred until after MVP is live and validated.

- AI movement analysis: user uploads sport video, AI detects posture issues, knee collapse, mobility restrictions
- AI injury risk screening: questionnaire produces personalised injury risk report
- Wearable integrations: Apple Watch heart rate + activity, Garmin training load
- Real-time multi-device sync
- Native iOS and Android apps
- Coach portal: trainers assign and review client plans
- B2B gym partnerships: white-label MotionLab dashboard for gym members
- Team features: group training, shared PRs, leaderboards within a squad
- Live session classes: coach-led group warmups or sport drills
- Pgvector semantic search for community and lesson discovery
- Marketplace: curated gear recommendations per sport with affiliate model

15. Open Questions

Question	Owner	Notes
Final brand name?	Both founders	MotionLab works but broader than either product alone. Decide before Phase 1 completes.
Freemium gate: what exactly is free vs paid?	Both founders	Recommend: 3 AI queries/day, 2 learning paths, community access = free. Unlimited AI, all paths, certifications = Pro.
Who owns content strategy for which sports?	Aditya + KavaFit founder	Aditya: TT and Football. Split remaining 6 sports by interest and expertise.
Expert compensation model beyond visibility?	Both founders	Revenue share on certified content once platform reaches ₹1L/month revenue.
Should deload detection be user-configurable?	KavaFit founder	Phase 3 item — current automatic logic is correct for MVP.
Progressive overload increments: configurable?	KavaFit founder	Phase 3 item — +5kg/+2.5kg defaults are fine for MVP.
Data retention policy for session_sets and photos?	Both founders	Required before GDPR/DPDP compliance work. Define before Phase 4.
USDA food fallback order for Indian foods not in USDA?	KavaFit founder	Currently falls back to indian_foods table. Confirm priority order.

Appendix: Key localStorage Keys

Key	Purpose	TTL
kavafit_active_session_v1	Current live session state (survives page refresh)	Until session end
kavafit_new_prs	Recent PRs	24 hours
kavafit_flags_*	AI insights cache (keyed by user ID)	24 hours
last_summary_date / last_weekly_summary	Weekly AI review — prevents duplicate generation	7 days
kavafit_gym_location	Last known location for gym finder	7 days
kavafit_gyms_cache	Gym list for last location	24 hours

MotionLab PRD — Version 1.0 — June 2026
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